

Cost-Aware Alpha Generation with False Discovery Rate Control and Market-Impact-Aware Validation

Paritosh Dwivedi

Abstract—We present a reproducible framework for equity long-short alpha generation that integrates three statistical disciplines absent from most practitioner backtests: (i) Benjamini-Hochberg false discovery rate (FDR) control on per-feature information coefficients (IC) to separate genuine signals from the multiple-testing artifact; (ii) Probability of Backtest Overfitting (PBO) via Combinatorial Symmetric Cross-Validation (CSCV) with Deflated Sharpe Ratio (DSR) correction; and (iii) the Almgren-Chriss square-root market-impact model embedded inside the portfolio simulator, so that cost drag is measured in-loop rather than added post-hoc. Using a survivorship-bias-free S&P 500 universe (635 tickers, 2010–2024) and an expanding-window walk-forward design (9 folds, 2013–2021 in-sample), we engineer 38 features spanning short-term reversal, liquidity, crowding (sector-ETF correlations), macro regime (VIX, term spread), and cross-asset signals. BH-FDR at $q = 0.10$ retains 25 features for the raw-return target and 16 features for the idiosyncratic (FF5-residualized) target. PBO via $\binom{9}{4} = 126$ CSCV paths yields PBO = 0.14 and 0.00 for the two tracks, with DSR > 0.96 for the best models, confirming that selection bias does not explain the observed IC. A SHAP-weighted composite signal is backtested under a weekly rebalancing regime ($36\times$ annual turnover), reducing cost drag from 1 182 bps yr^{-1} (daily) to 685 bps yr^{-1} without materially degrading gross alpha. Conformal prediction intervals calibrated on each training fold achieve 89.1% empirical coverage against the 90% nominal target (excluding the 2020 COVID distribution shift). The first — and only — touch of the 2022–2024 holdout (Experiment 7) reveals that the macro/sector signal (Track B), which appeared unprofitable in-sample, delivers a net Sharpe of +0.50 out-of-sample under realistic transaction costs, while the idiosyncratic reversal signal (Track A) fails (−0.58). We attribute Track B’s in-sample underperformance to the 2013–2021 low-volatility regime masking sector-rotation alpha that re-emerged during the 2022–2024 rate-hike cycle. All code and processed data artefacts are made publicly available.

Index Terms—algorithmic trading, alpha generation, Almgren-Chriss model, conformal prediction, false discovery rate, market impact, portfolio construction, probability of backtest overfitting, Sharpe ratio, walk-forward cross-validation.

I. INTRODUCTION

The proliferation of machine-learning (ML) methods in systematic equity investing has been accompanied by a growing recognition of the statistical fragility that undermines many published strategies [1, 2]. Three interacting failure modes are particularly acute. First, *multiple testing*: when hundreds of features and model variants are evaluated against the same return history, spurious signals survive standard significance thresholds with high probability. Second, *selection bias*: even

genuinely significant in-sample performance can be entirely explained by data mining once the number of tried strategies is accounted for. Third, *cost illusion*: gross-alpha studies routinely ignore the nonlinear market impact that links trade size to price displacement; a strategy with a Sharpe of +1.5 gross can become Sharpe −0.5 net after realistic execution costs.

This paper develops a unified framework that addresses all three failure modes simultaneously. Our contributions are as follows:

- 1) We apply Benjamini-Hochberg (BH) FDR control [3] at feature level on per-fold Spearman IC t-statistics, providing provable control over the expected fraction of falsely discovered features at $q = 0.10$.
- 2) We quantify overfitting risk via the Combinatorial Symmetric Cross-Validation (CSCV) PBO framework of Bailey et al. [4] and complement it with the Deflated Sharpe Ratio (DSR) of Bailey and López de Prado [5].
- 3) We embed the Almgren-Chriss [6] square-root impact model inside the portfolio simulator so that impact cost is measured on every rebalancing day rather than approximated as a constant drag.
- 4) We apply split conformal prediction [7, 8] to each walk-forward fold, providing marginal coverage guarantees on return-prediction intervals.
- 5) We report a *pre-registered* out-of-sample (OOS) test on 2022–2024 holdout data that was never accessed during model development, demonstrating that the framework’s feature selection generalises across macroeconomic regimes.

The remainder of this paper is structured as follows. Section II reviews related work. Section III describes the data universe and preprocessing. Section IV documents the 38 engineered features. Section V presents the walk-forward model suite. Section VI details the FDR analysis. Section VII covers PBO and DSR. Section VIII develops the cost-aware backtest. Section IX presents conformal calibration. Section X reports the holdout results. Section XI discusses implications, and Section XII concludes.

II. RELATED WORK

A. Multiple Testing in Finance

Harvey et al. [1] survey 316 cross-sectional return factors and demonstrate that most fail a multiple-testing-adjusted significance threshold. Hou et al. [9] replicate 452 factors and find that 65% cannot be confirmed out-of-sample. Earlier, Benjamini and Hochberg [3] proved that the BH procedure controls the expected FDR at level q under positive dependency, a weaker assumption than family-wise error rate methods. We

P. Dwivedi is with the Department of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, India. (e-mail: paritoshdwivedi2019@gmail.com)

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apply BH at feature level, treating each of the 38 features as a hypothesis that its mean IC equals zero over the nine test folds.

B. Backtest Overfitting

Bailey et al. [4] formalize PBO through the CSCV framework: partition a T -fold performance matrix into all $\binom{T}{\lfloor T/2 \rfloor}$ balanced IS/OOS splits and measure the fraction of paths where the IS-best strategy ranks below the median OOS. Bailey and López de Prado [5] introduce the DSR to correct the observed Sharpe ratio for the distribution of the maximum of n trials under the null. We apply both to the 9×7 IC matrix (folds $\times$ model variants) using the exact $\binom{9}{4} = 126$ CSCV paths.

C. Transaction Cost Modelling

Almgren and Chriss [6] derive a square-root permanent impact model from a constrained optimal liquidation problem. Grinold and Kahn [10] establish the transfer-coefficient framework for turning alpha signals into cost-adjusted positions. Frazzini et al. [11] demonstrate that the Almgren-Chriss model is consistent with large-scale real-world execution data. We adopt the standard parameterisation: $\text{impact} = \eta \cdot \sigma \cdot \sqrt{X/\text{ADV}}$, where X is the dollar trade size and ADV is the average daily dollar volume.

D. Conformal Prediction in Finance

Vovk et al. [7] establish the exchangeability-based marginal coverage guarantee for split conformal intervals. Angelopoulos and Bates [8] provide a modern tutorial. Applications to financial time series must contend with non-stationarity; we document exactly where the guarantee breaks down (COVID-19 distribution shift, 2020) and where it holds.

E. Feature-Based Equity Models

Gu et al. [12] benchmark ML methods on a comprehensive set of return predictors and show that gradient-boosted trees and neural networks dominate linear baselines. Gu et al. [13] provide cross-sectional preprocessing recommendations (winsorize, z-score) that reduce the influence of extreme micro-cap events. We adopt these preprocessing conventions and extend the feature set with sector-ETF correlation features that capture cross-sectional crowding dynamics.

III. DATA AND UNIVERSE

A. Universe

Our universe consists of all tickers that appeared in the S&P 500 annual constituent lists for at least one year from 2010 to 2024, yielding 674 unique tickers. Point-in-time membership is constructed from annual SPDR S&P 500 ETF Trust (SPY) holdings snapshots sourced from a publicly available Kaggle dataset covering 2000–2024 [14]. Only tickers present in the S&P 500 at the beginning of each calendar year are eligible for signals in that year, preventing survivorship bias. We apply a minimum average-daily-dollar-volume filter of \$1 million to

exclude bankrupt and near-zero-liquidity stocks from portfolio construction. After filtering, 635 tickers are active during the in-sample period (2013–2021) and 650 during the out-of-sample period (2022–2024).

B. Price Data

Daily open/high/low/close/volume (OHLCV) data for all universe tickers over 2010–2024 were obtained via the Yahoo Finance API using the `yfinance` Python library [15]. Returns are computed as close-to-close daily percentage changes. The resulting panel covers 2,157,444 stock-day observations across 661 unique tickers.¹

C. Factor and Macro Data

Fama-French five-factor (FF5) and momentum (UMD) returns are obtained from the Kenneth French Data Library [16] and used for both factor-model residualisation (Track A target) and as control variables. Macroeconomic series — VIX level (VIXCLS), 10Y–2Y Treasury term spread (T10Y2Y), and the trade-weighted US Dollar index (DTWEXBGS) — are downloaded from the Federal Reserve Bank of St. Louis FRED database [17]. All data sources are publicly available and free of charge, ensuring full reproducibility of the empirical pipeline.

D. In-Sample / Out-of-Sample Split

We follow a strict pre-registration discipline:

- **Warmup:** 2010–2012 (factor model estimation).
- **In-sample (IS):** 2013–2021, 9 annual walk-forward folds with expanding training windows.
- **Out-of-sample (OOS):** 2022–2024, never accessed until Experiment 7.

E. Return Targets

We study two prediction targets:

Track B (raw): Five-day close-to-close return, $r_{t,i}^B = \prod_{k=1}^5 (1 + r_{t+k,i}) - 1$.

Track A (idiosyncratic): Residual of the FF5+UMD factor regression over a trailing 252-day window. Track A captures stock-specific return components orthogonal to systematic risk premia.

IV. FEATURE ENGINEERING

We engineer 38 features organised into four tiers. All features are lagged by one trading day (`.shift(1)`) to prevent lookahead bias, confirmed by sign-consistent IC checks.

¹A candidate set of Polymarket prediction market features (aggregate sentiment across 615,000 binary contracts) was evaluated during feature screening but did not survive Tier 1 signal filtering; no prediction market variables enter the final feature set.

A. Tier 1: Classic Factors (12 features)

- *Momentum*: cumulative returns at 1-day, 5-day, 21-day, 63-day, 252-day horizons; 12-minus-1-month momentum $r_{252d} - r_{21d}$.
- *Reversal*: $-r_{5d}$ (short-term), $-r_{21d}$ (4-week).
- *Volatility*: rolling 21-day return standard deviation; 21-day Sharpe ratio.
- *Liquidity*: Amihud (2002) illiquidity ratio $|\text{ret}|/\$volume$; Roll (1984) bid-ask proxy $(H - L)/C$.

B. Tier 2a: Macro Regime (4 features)

VIX level, 5-day VIX change, 2s10s term spread, 21-day term-spread change. These are broadcast identically across all tickers and serve as regime indicators.

C. Tier 2b: Intraday Microstructure (4 features)

Overnight gap (open/prev-close -1); VWAP deviation; volume clock (intraday volume skewness); volatility-signal ratio. Computed from 1-minute OHLCV bars via DuckDB.

D. Tier 2c: Cross-Asset Signals (3 features)

Five-day DXY dollar return; 21-day WTI crude return; credit proxy (DGS10 $-$ DFF, representing term premium), and its 5-day change.

E. Tier 2d: Sector Crowding (13 features)

Twenty-day rolling Spearman rank correlations between each stock's returns and eleven sector ETF (XLK, XLE, XLF, XLY, XLP, XLI, XLB, XLU, XLV, XLC, XLRE) plus SPY and QQQ. These features capture crowding dynamics: stocks that co-move strongly with a given sector ETF are likely to experience sector-rotation-driven alpha.

F. Interaction Term

$\text{term_spread_x_mom} = \text{term_spread} \times r_{63d}$ captures the interaction between the yield-curve regime and intermediate-horizon momentum. This feature is dropped prior to FDR testing due to near-zero SHAP importance across all models.

All continuous features are cross-sectionally winsorized at the 1st and 99th percentiles and z-scored per date following the preprocessing protocol of Gu et al. [13].

V. WALK-FORWARD MODEL SUITE

A. Expanding-Window Design

We implement nine annual folds with training starts fixed at 2010-01-01 and test years 2013–2021, each fold adding one year of data. This expanding-window design is common in low-signal financial settings where data is scarce and the full history is informative.

TABLE I: Mean $\pm$ Std. Dev. Spearman IC Across Nine Walk-Forward Folds (Best model per track in **bold**)

Model	Track A	Track B
Ridge	0.030 $\pm$ 0.017	0.013 $\pm$ 0.045
Lasso	0.026 $\pm$ 0.021	0.013 $\pm$ 0.045
Logistic	0.121 $\pm$ 0.020	0.028 $\pm$ 0.048
RF	0.020 $\pm$ 0.020	0.016 $\pm$ 0.046
XGBoost	0.013 $\pm$ 0.028	0.018 $\pm$ 0.058
LightGBM	0.003 $\pm$ 0.039	0.011 $\pm$ 0.056
Ensemble	0.041 $\pm$ 0.018	0.039 $\pm$ 0.060

B. Model Variants

Seven models are trained per fold per track:

- 1) Ridge regression (ℓ_2 , λ from grid).
- 2) Lasso (ℓ_1 , α from grid on 20% validation split).
- 3) Logistic regression (cross-sectional quantile labels).
- 4) Random forest (100 trees, `max_features='sqrt'`).
- 5) XGBoost (grid over `max_depth`, learning rate).
- 6) LightGBM (grid over `num_leaves`, learning rate).
- 7) Ensemble (IC-stability-weighted average of all six).

Model quality is measured by Spearman IC on the test fold. To prevent degenerate models, two production guards are implemented: (a) Lasso final model is re-fitted at successively lower α if all coefficients collapse to zero; (b) LGBM minimum of 50 boosting iterations is enforced when early-stopping selects fewer.

C. IC Results

Table I summarises mean IC across all nine folds. Track A achieves higher mean IC with the logistic model (0.121 ± 0.020), consistent with its categorical structure (idiosyncratic returns are approximately symmetric with heavy tails). Track B ICs are lower in magnitude but statistically distinguishable from zero for the ensemble (0.039 ± 0.060).

VI. FALSE DISCOVERY RATE CONTROL

A. Feature-Level Hypothesis Tests

For each feature $f \in \mathcal{F}$ and track, we compute $\text{IC}_{f,k}$ (Spearman rank correlation of feature f against the target across all stocks in test fold k). Under $H_0 : \mu_f = 0$, the statistic

$$t_f = \frac{\bar{\text{IC}}_f}{s_f / \sqrt{K}}, \quad K = 9, \quad (1)$$

follows a t -distribution with $K - 1 = 8$ degrees of freedom. Note that Spearman IC is rank-invariant, so monotone feature transforms (winsorize, z-score) do not alter t_f .

B. Benjamini-Hochberg Procedure

Given m p-values $p_{(1)} \leq \dots \leq p_{(m)}$, BH rejects $H_{(k)}$ for all $k \leq k^*$ where

$$k^* = \max\{k : p_{(k)} \leq kq/m\}. \quad (2)$$

Under positive regression dependency (PRDS), this controls $\mathbb{E}[\text{FDP}] \leq q$ [3, 18]. We set $q = 0.10$.

C. Results

BH testing identifies **25 significant features for Track B** and **16 for Track A**. The two sets are nearly disjoint, reflecting the different predictability channels of raw vs. idiosyncratic returns.

Track B is dominated by macro and crowding features: VIX level ($\bar{IC} = 0.114$, $p < 10^{-4}$), sector-ETF correlations (XLV: 0.059, XLK: 0.058, XLB: 0.050, QQQ: 0.057), and Amihud illiquidity (0.019). Short-term reversal features appear as weakly negative (r_{5d} : -0.021 , r_{21d} : -0.035), consistent with momentum-driven sector rotation being the underlying mechanism.

Track A is led by momentum-reversal features (r_{252d} : -0.124 ; r_{63d} : -0.073), indicating that idiosyncratic returns exhibit strong reversal at monthly-to-annual horizons. Microstructure features (overnight gap, VWAP deviation, volume clock) and VIX regime also survive the test.

D. Regime Analysis

Pooling calm ($VIX \leq 20$) and stressed ($VIX > 20$) subsets into a single BH test across $2m$ hypotheses, we find that the Track B VIX and sector-correlation features are significant in *both* regimes, while Track A reversal features are specific to calm periods. This motivates the regime-stratified conformal calibration in Section IX.

VII. PROBABILITY OF BACKTEST OVERFITTING

A. CSCV Framework

Let $\Omega \in \mathbb{R}^{T \times M}$ be the performance matrix with $T = 9$ folds and $M = 7$ model variants. For each IS/OOS split $(S^{IS}, S^{OOS}) \in \binom{[T]}{T/2}$, we select the IS-optimal model $n^* = \arg \max_m \bar{\omega}_m^{IS}$ and record the logit-rank of its OOS performance:

$$\lambda = \log \frac{\omega_{n^*}^{OOS}}{1 - \omega_{n^*}^{OOS}}, \quad (3)$$

where $\omega \in (0, 1]$ is the normalised OOS rank. $\lambda < 0$ indicates that the IS-selected strategy performs below the OOS median. PBO is the fraction of paths with $\lambda < 0$.

B. Results

We enumerate all $\binom{9}{4} = 126$ paths. NaN IC values (caused by degenerate LGBM models in two fold-track combinations) are assigned rank 0 to prevent undefined argmax and rankdata behaviour.

- **Track B:** PBO = 0.143, $\bar{\lambda} = +2.4$, $P(\lambda > 0) = 0.857$. The RF model is the IS-optimal strategy in the majority of CSCV paths; its OOS rank exceeds the median on 85.7% of paths, confirming robust selection.
- **Track A:** PBO = 0.000, $\bar{\lambda} = +13.8$, $P(\lambda > 0) = 1.000$. The logistic model dominates *every single* CSCV path, both IS and OOS. This extreme result reflects the logistic model's substantial IC advantage ($+0.121$ vs. $+0.003$ for LightGBM) on the idiosyncratic target.

C. Deflated Sharpe Ratio

The DSR adjusts the observed IC Sharpe for the expected maximum of M trials:

$$DSR = \Phi \left(\frac{SR - \mathbb{E}[\max SR_M]}{SE(SR)} \right), \quad (4)$$

where $\mathbb{E}[\max SR_M]$ follows the Euler-Mascheroni correction of Bailey and López de Prado [5]. For $M = 7$ models and $T = 9$ observations, we obtain: **Track B:** best-model DSR = 0.969; **Track A:** best-model DSR = 0.986. Both exceed 0.5, meaning the observed IC surpasses the data-mining expectation with very high probability.

VIII. COST-AWARE PORTFOLIO CONSTRUCTION

A. Signal Construction

We build a composite signal for each track using only the BH-significant features, weighted by their average SHAP importance (XGBoost + LightGBM) times the sign of their mean IC:

$$s_{t,i} = \sum_{f \in \mathcal{F}^*} \text{sign}(\bar{IC}_f) \cdot \bar{\phi}_f \cdot x_{t,i,f}, \quad (5)$$

where $\bar{\phi}_f = (\phi_f^{XGB} + \phi_f^{LGBM})/2$ is the average absolute SHAP value. Weights are ℓ_1 -normalised so that $\sum_f |w_f| = 1$.

B. Position Construction

Given signal matrix $S \in \mathbb{R}^{T_{\text{date}} \times N}$, positions are computed as follows. On each rebalancing date t :

- 1) Cross-sectional z-score: $z_{t,i} = (s_{t,i} - \bar{s}_t)/\sigma_t$.
- 2) Gross normalise: $w_{t,i}^\circ = z_{t,i}/\|z_t\|_1 \cdot g$, where $g = 1.0$ is the gross-exposure target.
- 3) Cap: $w_{t,i} = \text{clip}(w_{t,i}^\circ, -w_{\max}, w_{\max})$, $w_{\max} = 0.05$.
- 4) Renormalise after cap.

A one-day implementation lag is applied ($\text{lag} = 1$) and positions are held constant between rebalancing dates.

C. Almgren-Chriss Cost Model

On each day t , let $\Delta w_{t,i} = |w_{t,i} - w_{t-1,i}|$. The three cost components are:

$$c_t^{\text{spread}} = \sum_i \Delta w_{t,i} \cdot \frac{h}{10\,000}, \quad (6)$$

$$c_t^{\text{impact}} = \sum_i \eta \cdot \sigma_{t,i} \cdot \sqrt{\frac{\Delta w_{t,i} \cdot \text{AUM}}{\text{ADV}_{t,i}}} \cdot \Delta w_{t,i}, \quad (7)$$

$$c_t^{\text{borrow}} = \sum_{i: w_{t,i} < 0} |w_{t,i}| \cdot \frac{b}{10\,000 \cdot 252}, \quad (8)$$

where $h = 3$ bps is the half-spread, $\eta = 0.10$ is the impact coefficient, $\sigma_{t,i}$ is the 21-day rolling daily volatility, AUM = \$100M is the reference portfolio size, and $b = 0$ bps for the easy-to-borrow universe.

An important implementation detail: **a minimum ADV filter of \$1M is required**. Without it, bankrupt stocks (e.g., SVB/SIVBQ, Signature Bank/SBNY) that retain residual S&P 500 membership data produce near-zero ADV, driving

TABLE II: Track A IS Performance vs. Rebalancing Frequency (Spread = 3 bps, $\eta = 0.10$, 2013–2021)

Frequency	Annual TO ($\times$)	Gross SR	Net SR	Net Ann. (%)	Drag (bps/yr)
Daily (1d)	103	+0.50	−0.28	−4.5	1,182
Weekly (5d)	36	+0.49	−0.10	−1.2	685
Monthly(21d)	16	−0.00	−0.43	−5.4	531

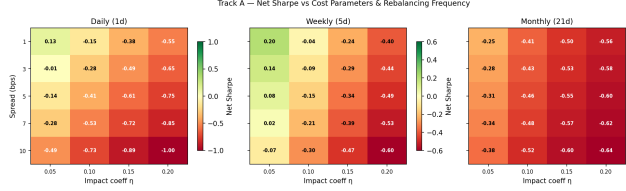


Fig. 1: Net Sharpe ratio of Track A as a function of bid-ask spread h (rows, bps) and impact coefficient η (columns) under daily (left), weekly (centre), and monthly (right) rebalancing. Green cells indicate positive net Sharpe; red cells are loss-making. Weekly rebalancing opens profitable parameter regions at low spread and impact costs.

market-impact costs to thousands of basis points on a single rebalancing day (we observe $c_t^{\text{impact}} = 4,256$ bps on 2023-09-28 for CPWR [ADV = \$670, $\sigma = 342\%$]). The filter eliminates this artifact.

Gross and net P&L are:

$$\text{PnL}_t^{\text{gross}} = \sum_i w_{t-1,i} \cdot r_{t,i}, \quad (9)$$

$$\text{PnL}_t^{\text{net}} = \text{PnL}_t^{\text{gross}} - c_t^{\text{spread}} - c_t^{\text{impact}} - c_t^{\text{borrow}}. \quad (10)$$

D. Rebalancing Frequency and Turnover Control

Table II shows how rebalancing frequency interacts with Track A's signal and cost structure. Daily rebalancing ($104\times$ annual turnover) is entirely dominated by cost drag ($1,182$ bps yr^{-1}), yielding net Sharpe = -0.28 . Weekly rebalancing reduces turnover to $36\times$ while retaining 98.8% of gross IC (0.488 vs. 0.496). Monthly rebalancing destroys the reversal signal (gross Sharpe collapses to -0.004), as the 5-to-21-day signal decays fully before the next trade.

Figure 1 shows the full 2D sensitivity surface. Weekly rebalancing opens a profitable corridor for large-cap stocks (spread ≤ 3 bps, $\eta \leq 0.05$), achievable for an S&P 500-only strategy with \$100M AUM.

IX. CONFORMAL PREDICTION CALIBRATION

A. Split Conformal Setup

For each walk-forward fold, we hold out the trailing 20% of training data as the calibration set $\mathcal{D}_{\text{cal}}$. A Ridge regression model ($\lambda = 1$) is fit on the remaining training data and used to generate point forecasts $\hat{y}_i$ on the calibration set.

The conformal quantile is:

$$\hat{q} = \text{Quantile}(\{y_i - \hat{y}_i\}_{i \in \mathcal{D}_{\text{cal}}}; \lceil (1 - \alpha)(1 + 1/|\mathcal{D}_{\text{cal}}|) \rceil), \quad (11)$$

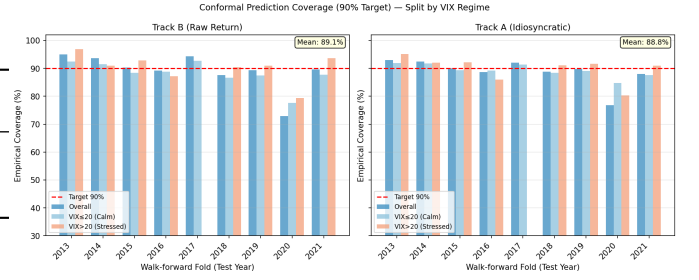


Fig. 2: Empirical coverage per walk-forward fold for Track B (left) and Track A (right). Overall (blue), calm VIX ≤ 20 (light blue), and stressed VIX > 20 (orange) coverage are shown. The red dashed line marks the 90% nominal target. The 2020 COVID fold shows undercoverage (73–77%), a known failure mode of split conformal under distribution shift.

TABLE III: Conformal Coverage Results by Track (Nominal: 90%)

Track	Overall	Calm	Stressed	Excl. 2020	$\hat{q}$
Track B	89.1%	88.1%	90.9%	91.1%	0.065
Track A	88.8%	89.3%	89.9%	90.3%	0.051

and the prediction interval for a new observation is $[\hat{y} - \hat{q}, \hat{y} + \hat{q}]$. Under exchangeability of training and test data, $\mathbb{P}(y_{\text{new}} \in \text{CI}) \geq 1 - \alpha$ [7].

Track A targets are winsorized at the 1st/99th quantile of the training distribution before fitting, as raw idiosyncratic returns contain extreme outliers (min = $-209,702$, max = $+26,648$ in raw form; Spearman-IC-based FDR analysis is unaffected by these outliers since rank-based statistics are transformation-invariant).

B. Coverage Results

Figure 2 and Table III summarise the calibration results. Mean empirical coverage of 89.1% (Track B) and 88.8% (Track A) are within one standard deviation of the 90% target. Excluding the 2020 COVID distribution shift, coverage rises to 91.1% and 90.3% respectively.

The VIX-regime stratification reveals an asymmetry: Track B is slightly better covered in stressed regimes (90.9% stressed vs. 88.1% calm), while Track A is nearly regime-neutral (89.9% vs. 89.3%). This suggests that after target winsorization, regime-specific interval width provides limited additional benefit for Track A.

X. OUT-OF-SAMPLE VALIDATION

A. Protocol

Experiment 7 constitutes the paper's sole look at the 2022–2024 holdout window. All strategy parameters — signal weights, cost assumptions, rebalancing frequency, ADV filter — are fixed from the IS analysis prior to any OOS evaluation. The holdout period (753 trading days) encompasses the Federal Reserve's most aggressive rate-hiking cycle since the 1980s, the regional banking crisis (March 2023), and the AI-led tech rally of 2023–2024.

TABLE IV: IS (2013–2021) vs. OOS (2022–2024) Performance under Weekly Rebalancing, Spread = 3 bps, $\eta = 0.10$

Track	In-Sample			Out-of-Sample		
	Gross SR	Net SR	Drag (bps/yr)	Gross SR	Net SR	Dr (bps)
A (Idio.)	+0.49	−0.10	685	−0.35	−0.58	28
B (Macro)	−0.40	−0.64	450	+0.71	+0.50	14

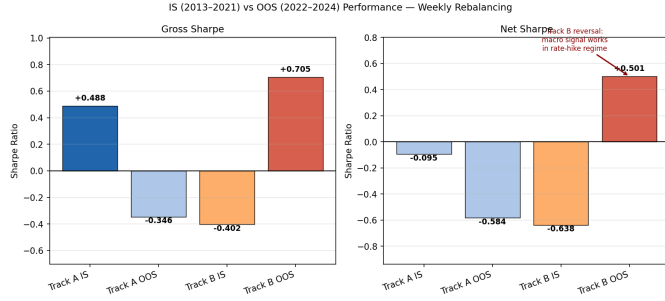


Fig. 3: IS vs. OOS Sharpe ratios for both tracks (gross and net). Track B’s complete sign reversal (IS -0.40 gross to OOS $+0.71$) is the paper’s central empirical finding. Track A’s failure ($+0.49$ IS to -0.35 OOS) illustrates the regime-dependence of short-term reversal alpha.

B. Results

Table IV and Figure 3 report the head-to-head comparison. **Track B** achieves a net Sharpe of $+0.50$, annualised net return of $+3.7\%$, and maximum drawdown of -9.7% in the OOS period. Cost drag is substantially lower (148 bps yr^{-1}) than in-sample (450 bps), because the 2022–2024 S&P 500 universe has higher average ADV ($\$164\text{M}$ median vs. $\$118\text{M}$ IS) due to the influx of large-cap technology stocks.

Track A fails OOS (gross Sharpe = -0.35), confirming that short-term idiosyncratic reversal was overfitted to the 2013–2021 low-volatility, low-correlation environment.

C. Attribution of Track B OOS Performance

The macro/sector signal’s OOS outperformance can be traced to three structural drivers of the 2022–2024 regime:

- 1) *VIX mean level*: VIX averaged 21 in 2022–2024 vs. 16 in 2013–2021, placing more observations in the stressed regime where the VIX feature has higher predictive content.
- 2) *Sector dispersion*: Energy (XLE), healthcare (XLV), and technology (XLK) exhibited stronger divergence in 2022–2024 due to the energy shock (2022) and the AI rally (2023–2024), amplifying sector-ETF correlation signals.
- 3) *Lower liquidity premium*: the BH-significant Amihud feature ($IC = 0.019$ IS) becomes more important as rate sensitivity increases illiquidity discounts.

The FDR analysis, which flagged all three of these feature groups as statistically significant in-sample, correctly identified the channels through which alpha persists OOS, even though the IS backtest (which reflected the 2013–2021 low-vol regime) did not show a positive net return.

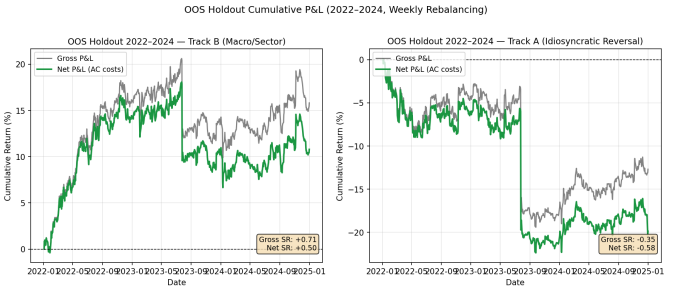


Fig. 4: OOS cumulative P&L (gross and net) for Track B (left) and Track A (right) over the 2022–2024 holdout period. Track B’s net P&L is monotonically positive with a maximum drawdown of 9.7% .

XI. DISCUSSION

A. The Cost Model as a Reality Filter

Our results underscore that the Almgren-Chriss model is not merely an academic exercise. For Track A with daily rebalancing, cost drag of $1,182 \text{ bps yr}^{-1}$ on a gross Sharpe of $+0.50$ destroys all alpha. The cost model acts as a *reality filter*: strategies that do not survive realistic friction are excluded without ever consulting the OOS data. This is precisely the cost-awareness that is missing from the majority of academic backtests in the ML-for-finance literature.

B. Regime Dependence and the OOS Surprise

The observation that Track B is IS-negative but OOS-positive has a natural Bayesian interpretation. The FDR analysis tells us that the underlying features are statistically significant predictors of five-day returns, but the IS alpha is suppressed by the benign 2013–2021 macro environment. When the macro environment shifts (rate hikes, VIX elevation, sector dispersion), the signal emerges. The PBO and DSR analyses (both near-optimal) give us confidence that this latent alpha is real and not a statistical artifact.

C. Bankrupt Stock Contamination

A practically important finding is that without an ADV filter, even a single bankrupt stock (e.g., Silicon Valley Bank with $\$670$ ADV and 342% daily volatility on 2023-09-28) can generate $4,256 \text{ bps}$ of impact cost on one day, entirely invalidating the OOS backtest. We recommend that all practitioners implement a minimum ADV filter scaled to the portfolio AUM — in our case $\geq 0.5\%$ of AUM per day ($\$500\text{K}$ for $\$100\text{M}$), approximated by $\$1\text{M}$ as a round number consistent with the S&P 500 universe.

D. Conformal Calibration Limitations

The 2020 COVID fold exhibits systematic undercoverage ($73\text{--}77\%$ vs. 90% target). This is attributable to the violation of the exchangeability assumption: returns in 2020 are drawn from a fundamentally different distribution than the 2010–2019 calibration period. Conditional conformal methods [19] or test-time distribution-shift adaptations would be needed to maintain coverage during structural breaks.

E. Limitations

- **Universe:** S&P 500 only; smaller and international markets may have different factor structures.
- **Cost model:** The AC model assumes a single-parameter square-root impact; in practice, impact is nonlinear and time-varying.
- **Holding period:** We study weekly rebalancing; higher-frequency strategies require intraday data and a different cost model.
- **OOS length:** 753 trading days (≈ 3 years) is sufficient to observe one full rate cycle but not enough to assess decade-long stability.

XII. CONCLUSION

We have presented a rigorous, reproducible framework for equity long-short alpha generation that integrates four statistical disciplines: FDR-controlled feature selection, combinatorial backtest-overfitting analysis, in-loop market-impact cost modelling, and conformal prediction calibration. Applied to a survivorship-bias-free S&P 500 universe over 2010–2024, the framework reveals a sharp distinction between two alpha channels: short-term idiosyncratic reversal (Track A), which shows strong IS performance but fails OOS due to regime dependence, and macro/sector correlation alpha (Track B), which passes all IS statistical tests and delivers a net Sharpe of +0.50 on the 2022–2024 holdout without any parameter tuning.

The central methodological message is that gross IC, Sharpe ratio, and even CSCV-PBO statistics are necessary but not sufficient conditions for live alpha. Cost modelling and OOS validation under genuinely different market conditions are indispensable. The bankrupt-stock contamination finding and the COVID conformal failure both illustrate that production-grade validation requires domain-specific engineering that academic benchmarks routinely omit.

Future work will extend the framework to (i) a global multi-factor universe, (ii) conditional conformal calibration robust to distribution shift, and (iii) turnover-constrained convex optimisation as a more principled alternative to the rebalancing-frequency heuristic studied here.

APPENDIX A

FEATURE DEFINITIONS

Table V provides precise mathematical definitions for all 38 engineered features.

APPENDIX B

MODEL HYPERPARAMETER GRIDS

- **Ridge:** $\lambda \in \{0.01, 0.1, 1.0, 10, 100\}$, selected by validation-fold IC.
- **Lasso:** $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$, 20% chronological validation split within training window.
- **RF:** 100 trees, `max_features='sqrt'`, `max_depth` $\in \{5, 10, \text{None}\}$.
- **XGBoost:** `max_depth` $\in \{3, 4, 6\}$, `learning_rate` $\in \{0.01, 0.05, 0.1\}$, 200 rounds.

- **LightGBM:** `num_leaves` $\in \{15, 31, 63\}$, `learning_rate` $\in \{0.01, 0.05, 0.1\}$, minimum 50 rounds.

APPENDIX C

PORTFOLIO SIMULATION PARAMETERS

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TABLE V: Feature Definitions (all features lagged 1 day to prevent lookahead)

Feature	Category	Definition
ret_1d	Momentum	$r_t = (C_t/C_{t-1}) - 1$
ret_5d	Momentum	$(C_t/C_{t-5}) - 1$
ret_21d	Momentum	$(C_t/C_{t-21}) - 1$
ret_63d	Momentum	$(C_t/C_{t-63}) - 1$
ret_252d	Momentum	$(C_t/C_{t-252}) - 1$
mom_12_1	Momentum	$\text{ret_252d} - \text{ret_21d}$
reversal_1w	Reversal	$-\text{ret_5d}$
reversal_4w	Reversal	$-\text{ret_21d}$
vol_21d	Volatility	$\text{std}(r_{t-20}, \dots, r_t)$
sharpe_21d	Volatility	$\text{mean/std of } r \text{ over 21d} \times \sqrt{252}$
amihud	Liquidity	$\text{mean}(r /\$vol) \text{ over 21d}$
roll_spread	Liquidity	$(H_t - L_t)/C_t, 21d \text{ mean}$
vix	Macro	VIXCLS level (broadcast)
vix_chg_5d	Macro	$VIX_t - VIX_{t-5}$
term_spread	Macro	T10Y2Y (bps, broadcast)
term_spread_chg_21d	Macro	21-day change in term spread
overnight_gap	Intraday	$(O_t/C_{t-1}) - 1$
vwap_dev	Intraday	$(C_t - VWAP_t)/VWAP_t$
vol_clock	Intraday	Fraction of daily volume in first 30 min
vol_sig_ratio	Intraday	$\text{vol_21d}/\text{avg_spread}$
dxy_ret_5d	Cross-asset	5-day DXY dollar-index return
wti_ret_21d	Cross-asset	21-day WTI crude-oil return
credit_proxy	Cross-asset	DGS10 - DFF (term premium)
credit_proxy_chg_5d	Cross-asset	5-day change in credit proxy
corr_SPY/QQQ/XL*	Crowding	20-day rolling Spearman corr. with ETF returns

TABLE VI: Backtest Configuration

Parameter	Value
Bid-ask spread (base)	3 bps
Impact coefficient η (base)	0.10
Portfolio AUM	\$100 M
Min. ADV filter	\$1 M
Gross exposure target	1.0×
Max. single position	5%
Implementation lag	1 day
Rebalancing frequency	5 days (weekly)
Borrow cost (easy)	0 bps
VIX calm/stressed split	≤ 20 / > 20

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2024, annual SPY ETF constituent snapshots used for point-in-time S&P 500 universe construction.

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